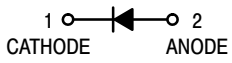
# Schottky Barrier Diode

# NSR05F30NXT5G

**30 V SCHOTTKY BARRIER DIODE**




These Schottky barrier diodes are optimized for low forward voltage drop and low leakage current and are offered in a Chip Scale Package (CSP) to reduce board space. The low thermal resistance enables designers to meet the challenging task of achieving higher efficiency and meeting reduced space requirements.

### Features

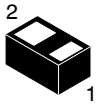
- Low Forward Voltage Drop – 400 mV @ 500 mA
- Low Reverse Current – 15 μA @ 10 V VR
- 500 mA of Continuous Forward Current
- ESD Rating – Human Body Model: Class 3B
  – Machine Model: Class C
- High Switching Speed
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant

### Typical Applications

- LCD and Keypad Backlighting
- Camera Photo Flash
- Buck and Boost dc−dc Converters
- Reverse Voltage and Current Protection
- Clamping & Protection

### Markets

- Mobile Handsets
- MP3 Players
- Digital Camera and Camcorders
- Notebook PCs & PDAs
- GPS




**DSN2**
**(0402)**
**CASE 152AC**

**MARKING DIAGRAM**




AN = Specific Device Code
M = Month Code

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| NSR05F30NXT5G | DSN2 (Pb−Free) | 5000 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

### MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Reverse Voltage | $V_R$ | 30 | V |
| Forward Current (DC) | $I_F$ | 500 | mA |
| Forward Surge Current (60 Hz @ 1 cycle) | $I_{FSM}$ | 10 | A |
| Repetitive Peak Forward Current (Pulse Wave = 1 sec, Duty Cycle = 66%) | $I_{FRM}$ | 4.0 | A |
| ESD Rating: Human Body Model<br>Machine Model | ESD | > 8<br>> 400 | kV<br>V |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

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## THERMAL CHARACTERISTICS

| Characteristic | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Thermal Resistance<br>Junction–to–Ambient (Note 1)<br>Total Power Dissipation @ $T_A$ = 25°C | <br>$R_{\theta JA}$<br>$P_D$ | | | <br>240<br>521 | <br>°C/W<br>mW |
| Thermal Resistance<br>Junction–to–Ambient (Note 2)<br>Total Power Dissipation @ $T_A$ = 25°C | <br>$R_{\theta JA}$<br>$P_D$ | | | <br>94<br>1.3 | <br>°C/W<br>W |
| Storage Temperature Range | $T_{stg}$ | | | –40 to +125 | °C |
| Junction Temperature | $T_J$ | | | +150 | °C |

1. Mounted onto a 4 in square FR–4 board 50 mm sq. 1 oz. Cu 0.06” thick single sided. Operating to steady state.
2. Mounted onto a 4 in square FR–4 board 1 in sq. 1 oz. Cu 0.06” thick single sided. Operating to steady state.

## ELECTRICAL CHARACTERISTICS ($T_A$ = 25°C unless otherwise noted)

| Characteristic | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Reverse Leakage<br>($V_R$ = 10 V)<br>($V_R$ = 30 V) | $I_R$ | | <br> | <br>15<br>75 | µA |
| Forward Voltage<br>($I_F$ = 100 mA)<br>($I_F$ = 500 mA) | $V_F$ | | <br>0.320<br>0.400 | <br>0.360<br>0.430 | V |



Share Feedback
Your Opinion Matters

## TYPICAL CHARACTERISTICS

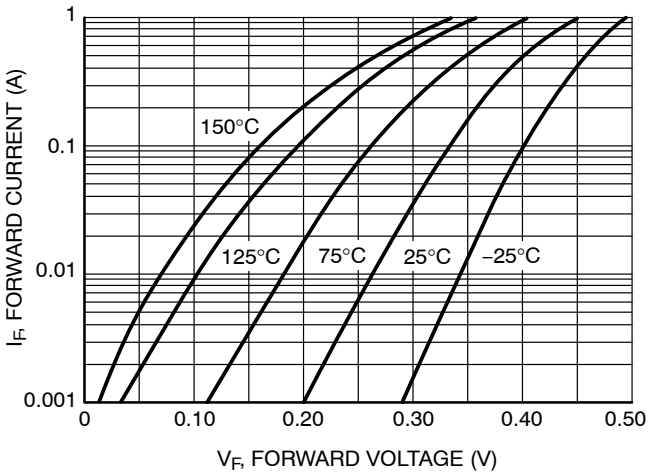



**Figure 1. Forward Voltage**

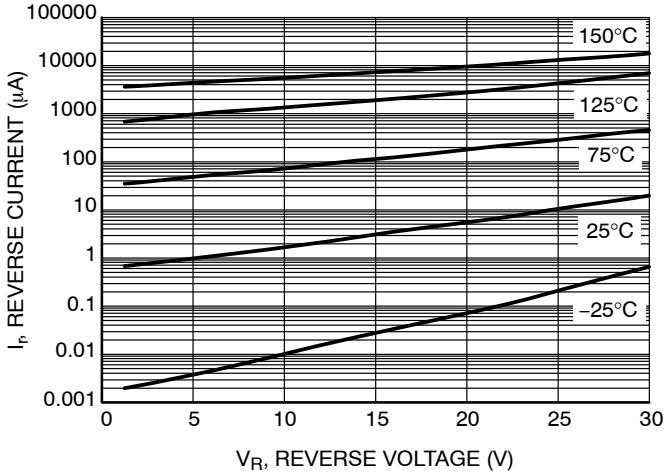



**Figure 2. Leakage Current**

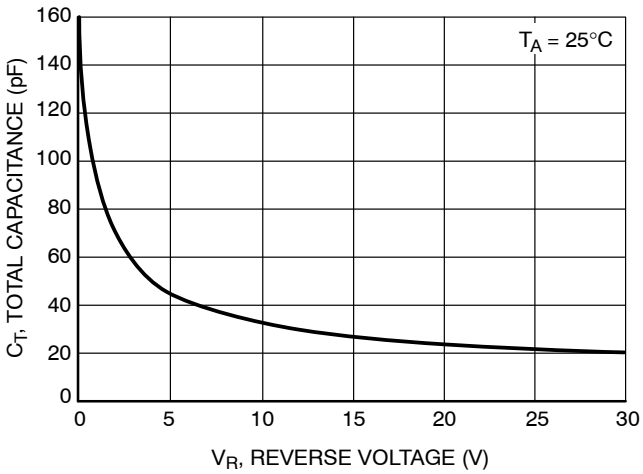



**Figure 3. Total Capacitance**

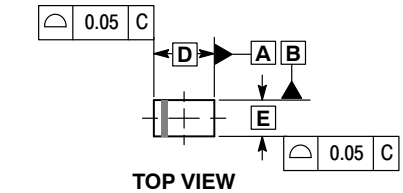


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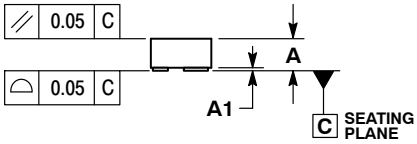
**DSN2, 1.0x0.6, 0.575P, (0402)**
CASE 152AC
ISSUE D

DATE 27 APR 2017

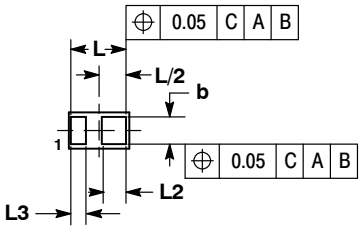
**SCALE 8:1**




**TOP VIEW**




**SIDE VIEW**

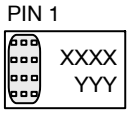



**BOTTOM VIEW**

NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: MILLIMETERS.

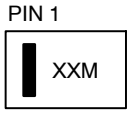
| | MILLIMETERS | |
|---|---|---|
| DIM | MIN | MAX |
| A | 0.25 | 0.31 |
| A1 | --- | 0.05 |
| b | 0.45 | 0.55 |
| D | 1.00 BSC | |
| E | 0.60 BSC | |
| L | 0.85 | 0.95 |
| L2 | 0.35 | 0.45 |
| L3 | 0.20 | 0.30 |

**GENERIC MARKING DIAGRAM1***




XXXX = Specific Device Code
YYY = Year Code

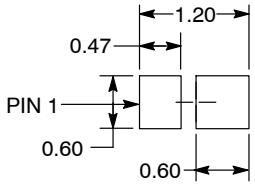
**GENERIC MARKING DIAGRAM2***




XX = Specific Device Code
M = Date Code

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G”, may or not be present. Some products may not follow the Generic Marking.

**RECOMMENDED SOLDER FOOTPRINT***

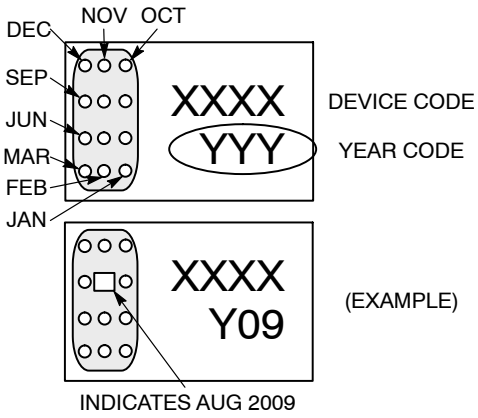



DIMENSIONS: MILLIMETERS

See Application Note AND8464/D for more mounting details

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

**CATHODE BAND MONTH CODING**




| | | | |
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| **DESCRIPTION:** | **DSN2, 1.0X0.6, 0.575P, (0402)** | | **PAGE 1 OF 1** |

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